

VN-Drive-JEPA: Causal Latent Predictive Architecture for Autonomous Vehicles in Vietnamese Traffic

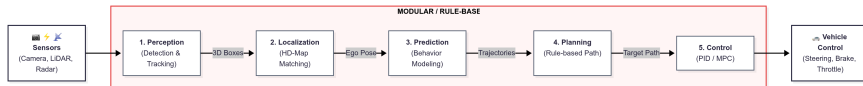
Latest update: September 17, 2026.

Authorized by Ngoc Ho

- ▶ Background
- ▶ Motivation and Problem Statement
- ▶ Related Work
- ▶ Proposed Method
- ▶ Conclusion

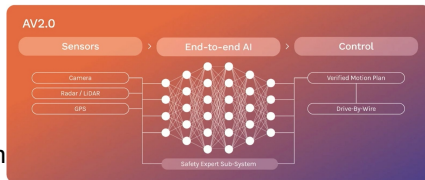
Background

- ▶ Autonomous driving is a major AI challenge, attracting substantial research effort and industrial investment.
- ▶ Traditional **modular, rule-based pipelines** decompose driving into sequential stages:



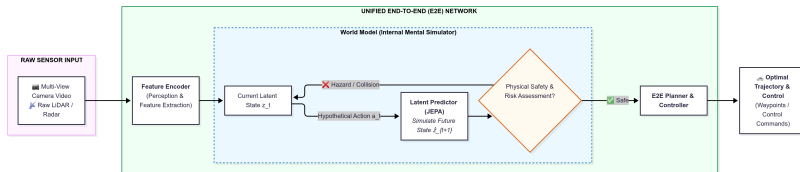
- ▶ Three core limitations follow from this sequential design:
 1. **Error propagation:** a small perception error can be amplified into an unsafe planning or control decision.
 2. **Information bottleneck:** hard labels such as 3D boxes and HD maps discard useful soft visual context.
 3. **Corner cases:** hand-written rules cannot cover the enormous variety of unexpected traffic situations.

- ▶ **End-to-end (E2E) driving** uses one unified neural network to map raw multi-view video, LiDAR, or radar directly to control commands or waypoints.
- ▶ Global end-to-end optimization replaces a chain of separate modules, preserving sensor context and reducing accumulated intermediate errors.
- ▶ **Challenge of reactive E2E:** it can map $O_t \rightarrow a_t$ as a black-box response, with limited explicit physical reasoning or long-horizon prediction.
- ▶ This can lead to unsafe decisions in rare events and complex, dynamic traffic scenes.



Source: [2]

- ▶ A **world model** predicts a future environmental state from the current state and a hypothetical action.
- ▶ It augments E2E with physical reasoning, longer-horizon prediction, and safer decision-making in complex traffic.



- ▶ As an **internal mental simulator**, it receives z_t and a_t to predict z_{t+1} in latent space.
- ▶ The planner rolls out multiple candidate actions, rejects predicted collision risks, and issues only a safer control action.

Two Paradigms of World Models for Autonomous Driving

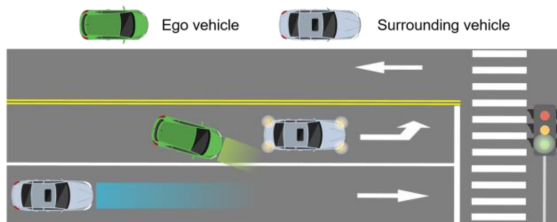


Criterion	Generative / Pixel-Based GAIA-1, DriveWM, DriveFuture	Latent World Model / JEPA V-JEPA, ACT-JEPA, LeWorldModel
Mechanism	Generates future RGB frames, often with autoregressive or diffusion models.	Predicts future semantic features directly in latent space, without decoding images.
Speed / cost	Requires heavy computation and may be too slow for real-time control.	Uses lightweight prediction, enabling fast closed-loop planning rollouts.
Data focus	Spends capacity on visual details such as textures, shadows, weather, and background.	Prioritizes task-relevant objects, motion, road geometry, and lane structure.
Main risk	Small pixel errors accumulate over time and can create distorted or hallucinated objects.	Latent drift or representation collapse may occur without careful loss and architecture design.

Motivation and Problem Statement

Problem 1: Motion Blind Spots in Static ViTs

- ▶ Static ViT backbones such as V-JEPA capture strong global semantics, but their attention is weak for small, fast-moving objects near the image boundary.
- ▶ In Vietnamese urban traffic, a motorcycle or scooter may occupy only a tiny image region and change position sharply across consecutive frames.
- ▶ This creates latent drift and unreliable collision-risk estimates when the model must reason about high-momentum agents.
- ▶ **Question:** how to inject motion information without destroying pretrained static representations?

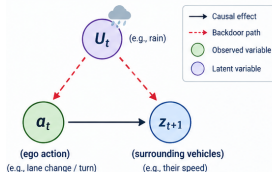


Source: [8]

- ▶ The ego action A_t and the hidden environment state U_t (weather, occlusion, nearby drivers' intent) jointly influence the future state S_{t+1} .
- ▶ If not modeled explicitly, the backdoor path $A_t \leftarrow U_t \rightarrow S_{t+1}$ induces spurious correlations in latent space.
- ▶ The predictor may mistake environmental changes for the effect of the ego action, leading to wrong planning decisions.

Problem 2: Causal Leakage in Action-Conditioned Prediction (cont'd.)

1. Causal graph with backdoor path

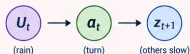


2. Real-world scenario



3. Model learns a spurious correlation

What the model observes



The model learns a (wrong) correlation:
"The ego car's turn causes other vehicles to slow down."

4. When intervening: $do(a_t)$

We ask: "What if the ego car turns ($do(a_t)$) while the rain does not happen?"



❌ Model prediction (wrong):
Other vehicles slow down (because it learned the spurious correlation).

5. Consequence

⚠ Prediction fails

The model confuses the driver's action with the effect of the weather, leading to incorrect future predictions ("hallucination").



► **Question:** how to prevent causal leakage from the hidden environment state U_t when predicting the effect of the ego action A_t ?

Problem 3: Closed-Loop Evaluation Without Dense Labels

- ▶ Standard closed-loop evaluation usually relies on CARLA/NAVSIM, HD maps, and expensive 3D annotations, which are not available for raw Vietnamese driving videos.
- ▶ This limits direct deployment of latent world-model planners on real-world data collected in dense, ambiguous urban scenes.



Source: [14]

- ▶ We therefore need a label-free latent-space evaluation: simulate actions, estimate risk, and reject unsafe trajectories without reconstructing the full 3D world.
- ▶ **Question:** how to evaluate the safety of latent-space rollouts without relying on dense 3D labels?

Related Work

JEPAs and World Models

Method	Mechanism	Strength	Limitation
V-JEPA (Bardes et al., 2024)	Self-supervised video representation learning with spatiotemporal masking.	Strong semantic features without a pixel decoder.	Not action-conditioned; static ViT can miss small, fast objects.
ACT-JEPA / World Models (LeCun, 2022)	Predicts future latent states conditioned on action a_t ; plans with CEM.	Estimates the effect of actions on z_{t+1} .	High planning latency; hidden environmental noise is not explicitly modeled.
GAIA-1 / UniAD (Wayve, 2023; Hu et al., 2023)	Generative future video or end-to-end perception-to-planning.	Rich visual generation or integrated driving tasks.	High compute cost or strong dependence on 3D boxes and HD maps.

Research Gap

No JEPA architecture directly integrates optical flow into the encoder to detect high-momentum objects near the vehicle.

Causal World Models

Method	Causal approach	Strength	Limitation
Causal World Models (Zhang et al., 2020)	Removes physical nuisance factors through latent deconfounding.	Demonstrates causal dynamics under controlled conditions.	Evaluated mainly in simple 2D environments; limited sensor complexity.
Causal RL (Pearl; Zhang et al., 2023)	Uses do-calculus and backdoor criteria for sequential decisions.	Provides a rigorous theory for reducing spurious correlations.	Requires a known causal graph or observable confounders.
Invariant Causal Prediction (Schölkopf et al., 2021)	Learns representations stable across domains.	Improves robustness under environment shifts.	Does not provide action interventions or latent rollouts.

Research Gap

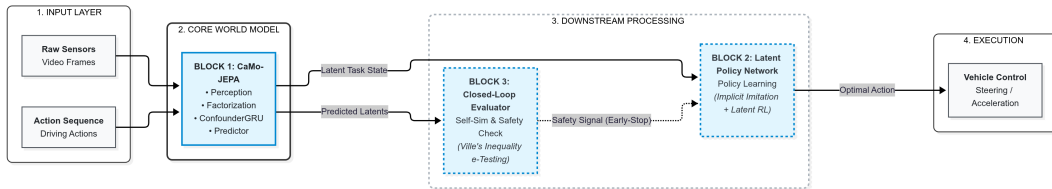
Existing methods do not orthogonally separate action-relevant z_{task} from exogenous z_{exo} while estimating the hidden confounder U_t in driving videos.

Method	Planning mechanism	Strength	Limitation
PrismAD (Li et al., 2026)	Separates feature streams for trajectory planning.	Improves planning in complex urban scenes.	Relies heavily on 3D boxes and HD maps.
DriveFuture / Diffusion (Li et al., 2026)	Generates future trajectories through iterative denoising.	Produces smooth and diverse trajectories.	Slow inference prevents real-time deployment.
TD-JEPA / DreamerV3 (Hafner et al., 2023)	Combines temporal-difference learning with latent world-model rollouts.	Supports long-horizon and zero-shot policy learning.	Lacks formal safety constraints; latent drift can accumulate.

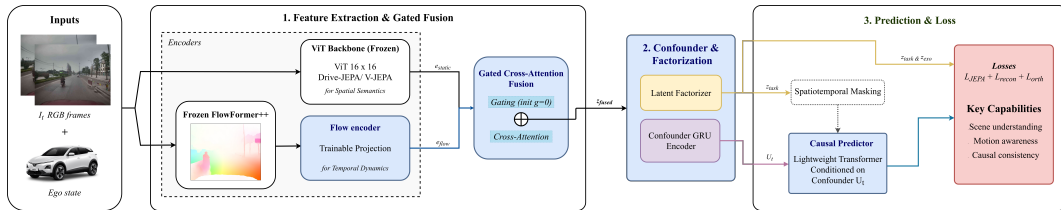
Research Gap

No method jointly learns from raw unlabeled sensors, evaluates closed-loop rollouts in latent space, and applies Ville's inequality for safe early stopping.

Proposed Method



- ▶ **Block 1 – CaMo-JEPA**: Causal Motion-Aware Joint Embedding Predictive Architecture for Self-Supervised Driving Perception: encodes raw video and action history, fuses static and motion features, and produces the causal latent state and future latent predictions.
- ▶ **Block 2 – Latent Policy Network**: uses the task-relevant latent state to generate candidate steering and acceleration actions.
- ▶ **Block 3 – Closed-Loop Evaluator**: rolls out each candidate in latent space, checks safety, and sends an early-stop signal before the optimal action is executed.



- ▶ **Input:** multi-view frames x_t and optical flow f_t .
- ▶ **Feature extraction:** Frozen Drive-JEPA ViT-L plus FlowTokenEncoder.
- ▶ **Latent processing:** Gated Fusion, QR factorization, and ConfounderGRU.
- ▶ **Action-conditioned predictor:** receives $z_{\text{task},t}$, the intervention $do(a_t)$, and hidden context U_t to predict future task states:

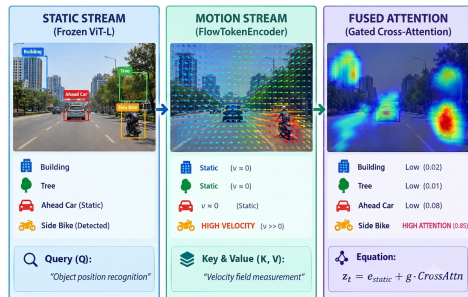
$$\hat{z}_{\text{task},t+1:t+H} = P_{\phi}(z_{\text{task},t}, do(a_t), U_t).$$
- ▶ **Output:** clean task state z_{task} , exogenous context z_{exo} , and predicted latent trajectory for planning and safety evaluation.

$$Q = e_{\text{static}} W_Q,$$

$$K = e_{\text{flow}} W_K,$$

$$V = e_{\text{flow}} W_V,$$

$$z_t = e_{\text{static}} + g \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V.$$



- ▶ e_{static} : broad semantic context from Frozen ViT-L.
- ▶ e_{flow} : local velocity cues from consecutive frames.
- ▶ $g = 0$ at initialization preserves the pretrained representation.

Purpose

Inject motion information into relevant tokens without overwriting static semantic features.

- ▶ Factorize the fused representation into action-relevant and exogenous components:

$$z_t = z_{\text{task},t} + z_{\text{exo},t}.$$

- ▶ Use QR-based projections to enforce orthogonality:

$$z_{\text{task}} = P_{\text{task}} z_t, \quad z_{\text{exo}} = P_{\text{exo}} z_t, \quad P_{\text{task}} P_{\text{exo}}^T = 0.$$

- ▶ The task branch receives action-related gradients; the exogenous branch preserves environmental context.
- ▶ This reduces leakage when the predictor applies $do(a_t)$.

Example: Causal Latent Factorization

Input tensor

$$\mathbf{z}_{\text{fused}} = \begin{bmatrix} 4 & 3 \end{bmatrix}$$

- ▶ Horizontal axis: exogenous context, e.g. weather or road condition.
- ▶ Vertical axis: task information, e.g. steering and braking state.

Step 1: orthogonal projections

$$\mathbf{P}_{\text{task}} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{P}_{\text{exo}} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

Step 2: project the input

$$\mathbf{z}_{\text{task}} = \mathbf{P}_{\text{task}} \mathbf{z}_{\text{fused}} = \begin{bmatrix} 0 & 3 \end{bmatrix},$$

$$\mathbf{z}_{\text{exo}} = \mathbf{P}_{\text{exo}} \mathbf{z}_{\text{fused}} = \begin{bmatrix} 4 & 0 \end{bmatrix}.$$

Step 3: verify the factorization

Reconstruction: $\mathbf{z}_{\text{task}} + \mathbf{z}_{\text{exo}} = \mathbf{z}_{\text{fused}}$

Orthogonality: $\mathbf{z}_{\text{task}} \mathbf{z}_{\text{exo}}^T = 0$

Step 4: causal intervention

If steering changes $\mathbf{z}_{\text{fused}}$ from $\begin{bmatrix} 4, 3 \end{bmatrix}$ to $\begin{bmatrix} 4, 8 \end{bmatrix}$, then $\mathbf{z}_{\text{task}} = \begin{bmatrix} 0, 8 \end{bmatrix}$ changes while $\mathbf{z}_{\text{exo}} = \begin{bmatrix} 4, 0 \end{bmatrix}$ remains fixed.

$$\begin{aligned}x_t &= \text{MeanPool}_{\text{patch}}(z_{\text{fused},t}), \\h_t &= \text{GRU}(x_t, h_{t-1}), \\U_t &= h_t, \quad U_t \in \mathbb{R}^{128}.\end{aligned}$$

Backdoor path

$$a_t \leftarrow U_t \rightarrow z_{t+1}$$

- ▶ Encodes slow-varying factors such as weather, occlusion, and surrounding-driver intent.
- ▶ Uses temporal history because a single frame cannot identify hidden environmental causes.
- ▶ Conditions the predictor without replacing z_{task} .

Example: ConfounderGRU

Input tensor

$B = 1$, two frames, three patches, four features:
 $\text{fused_patches} \in \mathbb{R}^{1 \times 2 \times 3 \times 4}$

Feature order:

[building/light, position, wetness, traffic/brake].

$t - 1$: three input patches

$P_1 = [0.9, 0.8, 0.1, 0.0]$	building
$P_2 = [0.1, 0.2, 0.8, 0.2]$	wet road
$P_3 = [0.2, 0.5, 0.3, 0.9]$	braking car

t : updated patches

$P_1 = [0.8, 0.9, 0.1, 0.0]$	building
$P_2 = [0.1, 0.1, 0.9, 0.3]$	wet road
$P_3 = [0.2, 0.4, 0.4, 0.9]$	braking car

Step 1: spatial mean-pooling

$$x_1 = [0.40, 0.50, 0.40, 0.37], \quad x_2 = [0.37, 0.47, 0.47, 0.40].$$

Step 2: reduced GRU

$$W_x = \begin{bmatrix} 0 & 0 & 0.7 & 0.7 \\ 0 & 0 & 0.4 & 0.6 \end{bmatrix}, \quad W_h = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \end{bmatrix}, \quad h_0 = [0, 0].$$
$$h_t = \tanh(W_x x_t + W_h h_{t-1}),$$
$$h_1 = \tanh([0.539, 0.382]) \approx [0.492, 0.364].$$

Step 3: hidden confounder output

$$h_2 = \tanh([0.609, 0.428] + [0.246, 0.182]) \approx [0.694, 0.544].$$
$$U_t = h_2 \in \mathbb{R}^2.$$

U_t accumulates persistent wet-road and braking context; building features are ignored because their input weights are zero.

Action-Conditioned SCM Predictor

- ▶ The causal predictor receives the clean task state, intervention, and hidden context:

$$\hat{z}_{\text{task},t+1} = P_{\phi}(z_{\text{task},t}, do(a_t), U_t).$$

- ▶ $do(a_t)$ changes only the action-relevant branch; z_{exo} is held stable during a rollout.
- ▶ Multiple candidate actions can therefore share the same environmental context and avoid repeated computation.
- ▶ The output is a long-horizon latent trajectory for policy selection and safety evaluation.

Key benefit

Causal conditioning separates the effect of the ego action from changes caused by the environment.

$$\mathcal{L}_{\text{total}} = \lambda_{\text{jepa}} \mathcal{L}_{\text{jepa}} + \lambda_{\text{orth}} \mathcal{L}_{\text{orth}} + \lambda_{\text{recon}} \mathcal{L}_{\text{recon}},$$

$$\mathcal{L}_{\text{jepa}} = \text{SmoothL1}(\text{LN}(z_{\text{pred}}), \text{LN}(z_{\text{target}})),$$

$$\mathcal{L}_{\text{orth}} = \left\| \frac{\bar{z}_{\text{task}}^T \bar{z}_{\text{exo}}}{N} \right\|_F^2,$$

$$\mathcal{L}_{\text{recon}} = 1 - \cos(z_{\text{fused}}, e_{\text{static}}).$$

- ▶ $\mathcal{L}_{\text{jepa}}$: predicts normalized target tokens with Smooth L1.
- ▶ $\mathcal{L}_{\text{orth}}$: decorrelates task and exogenous representations.
- ▶ $\mathcal{L}_{\text{recon}}$: keeps motion fusion close to static semantics.
- ▶ $\lambda_{\text{jepa}}, \lambda_{\text{orth}}, \lambda_{\text{recon}}$: configurable loss weights.

Variant	ViT-L	Flow	Factorization	ConfounderGRU
V1: V-JEPA baseline	Yes	No	No	No
V2: + Motion stream	Yes	Yes	No	No
V3: + Orthogonal loss	Yes	Yes	Yes	No
V4: + ConfounderGRU	Yes	Yes	No	Yes
V5: Full CaMo-JEPA	Yes	Yes	Yes	Yes

- ▶ Evaluate latent MSE, motion recall near the vehicle, and causal deconfounding error.
- ▶ Compare the independent contribution of motion, factorization, and hidden-confounder estimation.
- ▶ The full model combines all three mechanisms for robust long-horizon rollouts.

Conclusion

1. We propose **CaMo-JEPA**, which fuses Frozen Drive-JEPA ViT-L features with optical-flow features through gated cross-attention.
2. We introduce causal latent factorization with QR-based orthogonal projections and ConfounderGRU to separate task-relevant dynamics from exogenous context.
3. We formulate a three-term training objective combining JEPA prediction, orthogonality, and latent reconstruction losses.
4. We outline a label-free latent-space planning and evaluation framework with candidate-action rollouts and Ville-inequality-based early stopping.

Current scope

The CaMo-JEPA core world model is the focus of this work; the policy and closed-loop evaluator remain under development.

- [1] Chen, L., Wu, P., Chitta, K., Jaeger, B., Geiger, A., & Li, H. (2024). End-to-end autonomous driving: Challenges and frontiers. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(12), 10164–10183.
- [2] Lee, T. B. (2025). *Waymo and Tesla's self-driving systems are more similar than people think*. Understanding AI. <https://www.understandingai.org/p/waymo-and-teslas-self-driving-systems> (Accessed: 17/09/2026).
- [3] Guan, Y., Liao, H., Li, Z., Hu, J., Yuan, R., Zhang, G., & Xu, C. (2024). World models for autonomous driving: An initial survey. *IEEE Transactions on Intelligent Vehicles*.
- [4] LeCun, Y. (2022). A path towards autonomous machine intelligence (version 0.9.2). *OpenReview*.
- [5] Bardes, A., Garrido, Q., Ponce, J., Chen, X., Rabbat, M., LeCun, Y., ... & Ballas, N. (2024). V-JEPA: Video joint embedding predictive architecture. *arXiv preprint arXiv:2404.08471*.

- [6] Assran, M., Duval, Q., Misra, I., Bojanowski, P., Vincent, P., Rabbat, M., ... & Ballas, N. (2023). Self-supervised learning from images with a joint-embedding predictive architecture. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 15619–15629).
- [7] Hu, Y., Yang, J., Chen, L., Li, K., Sima, C., Zhu, X., ... & Li, H. (2023). Planning-oriented autonomous driving. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 17853–17862).
- [8] Mu, R., Yu, W., Li, Z., Wang, C., Zhao, G., Zhou, W., & Ma, M. (2024). Motion planning for autonomous vehicles in unanticipated obstacle scenarios at intersections based on artificial potential field. *Applied Sciences*, 14(4), 1626.
- [9] Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., ... & Houlsby, N. (2020). An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*.

- [10] Ranftl, R., Bochkovskiy, A., & Koltun, V. (2021). Vision transformers for dense prediction. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)* (pp. 12159–12168).
- [11] Pearl, J. (2009). *Causality: Models, Reasoning, and Inference* (2nd ed.). Cambridge University Press.
- [12] Schölkopf, B., Locatello, F., Bauer, S., Ke, N. R., Kalchbrenner, N., Goyal, A., & Bengio, Y. (2021). Toward causal representation learning. *Proceedings of the IEEE*, 109(5), 612–634.
- [13] Zeng, Y., Cai, R., Sun, F., Huang, L., & Hao, Z. (2024). A survey on causal reinforcement learning. *IEEE Transactions on Neural Networks and Learning Systems*, 36(4), 5942–5962.
- [14] CARLA Simulator. (2020). *CARLA Release 0.9.8*.
<https://carla.org/2020/03/09/release-0.9.8/>

- [15] Dosovitskiy, A., Ros, G., Codevilla, F., Lopez, A., & Koltun, V. (2017). CARLA: An open urban driving simulator. In *Conference on Robot Learning (CoRL)* (pp. 1–16). PMLR.
- [16] Dauner, D., Hallgarten, M., Li, T., Weng, X., Huang, Z., Yang, Z., ... & Chitta, K. (2024). Navsim: Data-driven non-reactive autonomous vehicle simulation and benchmarking. *Advances in Neural Information Processing Systems (NeurIPS)*, 37, 28706–28719.
- [17] Teed, Z., & Deng, J. (2020, August). Raft: Recurrent all-pairs field transforms for optical flow. In *European Conference on Computer Vision* (pp. 402–419). Springer, Cham.
- [18] Li, M., Yang, M., Liu, F., Chen, X., Chen, Z., & Wang, J. (2020). Causal world models by unsupervised deconfounding of physical dynamics. *arXiv preprint arXiv:2012.14228*.

- [19] Ding, K., Lin, Z., Wang, H., Gui, J., Liu, Q., Wang, Z., ... & He, L. (2026). PrismAD: Decoupled Planning via Semantic Mixture-of-Planners for End-to-End Autonomous Driving. *arXiv preprint arXiv:2607.10336*.
- [20] Li, X., Wang, B., Zhang, J., & Wang, J. (2026, June). Temporally Decoupled Diffusion Planning for Autonomous Driving. In *Proceedings of the International Conference on Automated Planning and Scheduling* (Vol. 36, No. 1, pp. 719–728).
- [21] Hafner, D., Pasukonis, J., Ba, J., & Lillicrap, T. (2023). Mastering diverse domains through world models. *arXiv preprint arXiv:2301.04104*.
- [22] Wang, J. (2026, March). Latentvla: Taming latent space for generalizable and long-horizon bimanual manipulation. In *Proceedings of the AAAI Conference on Artificial Intelligence* (Vol. 40, No. 22, pp. 18593–18601).
- [23] Ramdas, A., Grünwald, P., Vovk, V., & Shafer, G. (2023). Game-theoretic statistics and safe anytime-valid inference. *Statistical Science*, 38(4), 576–601.